

ENEL700 Tutorial 3

1. An AM wave displayed on an oscilloscope has values of $V_{\max}=4.8$ and $V_{\min}=2.5$ as read from the graticule. What is the percentage of modulation?
2. An AM radio transmitter operating on 3.9MHz is modulated by frequencies up to 4kHz. What are the maximum upper and lower side frequencies? What is the total bandwidth of the AM signal?
3. How much power appears in one sideband of an AM signal of a 5kW transmitter modulated by 80 percent?
4. The antenna current produced by an unmodulated carrier is 2.4A into an antenna with a resistance of 75Ω . When amplitude modulated, the antenna current rises to 2.7A. What is the percentage of modulation?
5. The total power content of an AM wave is 600 W. Determine the modulation index of the signal if each of the sidebands contains 75 W.